

24 number of turns

$$N = \frac{\text{tube length}}{\text{wire diameter}} = \frac{200}{0.050} = 4000$$

$$n = \frac{N}{\text{tube length}} = \frac{4000}{200 \times 10^{-3}} = 20\,000$$

$$\begin{aligned} R &= (8.66)(\text{wire length}) \\ &= (8.66)(N)(\text{circumference}) \\ &= (8.66)(N)(\pi)(\text{diameter}) \\ &= (8.66)(4000)(\pi)(30 \times 10^{-3}) \end{aligned}$$

$$\begin{aligned} B &= \mu_0 n I = \mu_0 n \frac{V}{R} \\ &= (4\pi \times 10^{-7})(20000) \frac{6.0}{(8.66)(4000)(\pi)(30 \times 10^{-3})} \\ &= 4.6 \times 10^{-5} \text{ T} \end{aligned}$$

25 By right hand grip rule, magnetic field points into paper.

magnetic force provides centripetal force

$$\begin{aligned} Bqv &= \frac{mv^2}{r} \\ r &= \frac{mv}{Bq} = \frac{(9.11 \times 10^{-31})(4.5 \times 10^6)}{(0.012)(1.6 \times 10^{-19})} \\ &= 2.1 \text{ mm} \end{aligned}$$